CMR UNIVERSITY

Private University Estd in Karnataka State by Act No. 45 of 2013

SCHOOL OF ENGINEERING AND TECHNOLOGY

A Project Report

On

"SMART HELMET SAFETY SYSTEM WITH ACCIDENT AND ALCOHOL DETECTION"

Submitted in partial fulfillment of the requirements for the award of degree in

BACHELOR OF TECHNOLOGY

IN

AIML

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Chagalahatti, Bengaluru, Karnataka-562149

Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the Project entitled "Smart Helmet Safety System with Accident and Alcohol Detection "has been successfully carried out by Prajwal Gowda C (Reg. No. 24BBTCA086), Prajwal Gowda J (Reg. No. 24BBTCA087), Surya Pratap (Reg. No. 24BBTCA092), and Sanjay Kumar (Reg. No. 24BBTCA106), in partial fulfillment of the requirement for the award of the degree **Bachelor of Technology in AIML of CMR University,** Bengaluru during the academic year **2025-2026.**The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide	**Signature of HOD**
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Sipla Sivan P	**Dr.K.V. PRASAD**
Assistant Professor	Head of the Department
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Examiners Signature with date:

1.

2.

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of this project would be incomplete without the mention of the people who made it possible, without whose constant guidance and encouragement would have made efforts go in vain.

I would like to express my thanks to **Dr.K.V. PRASAD, Professor and Head**, Department of Electronics and Communication Engineering, School of Engineering and Technology, CMR University, Bangalore, for his encouragement that motivated me for the successful completion of Project work.

I express my thanks to my Internal Project Guide Ms.Silpa Sivan P**, Assistant Professor**, Department of Electronics and communication Engineering, School of Engineering and Technology, CMR University for her constant support.

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DECLARATION

We, Prajwal Gowda C (Reg. No. 24BBTCA086), Prajwal Gowda J (Reg. No. 24BBTCA087), Surya Pratap (Reg. No. 24BBTCA092), and Sanjay Kumar (Reg. No. 24BBTCA106), students of IV Semester B.Tech, Artificial Intelligence and Machine Learning, School of Engineering and Technology, CMR University, Bangalore, hereby declare that the project work entitled "Smart Helmet Safety System" has been carried out by us under the guidance of Ms. Silpa Sivan P, Assistant Professor, Department of Artificial Intelligence and Machine Learning, School of Engineering and Technology.

This report is submitted in partial fulfillment of the requirements for the award of Bachelor of Technology in AIML by CMR University, Bangalore, during the academic year 2025-2026. The project report has been prepared by us and has not been submitted elsewhere for the award of any degree.

Place: Bangalore

Date:

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ABSTRACT

The Smart Helmet Safety System is designed to improve the safety of two-wheeler riders by ensuring proper helmet usage and preventing unsafe driving conditions. This project integrates multiple sensors such as an IR sensor for helmet detection, an MQ-3 alcohol sensor for detecting alcohol consumption, and a vibration sensor for accident detection. These sensors are connected to an Arduino Uno, which acts as the main controller.

The system checks whether the rider is wearing a helmet and ensures that the rider has not consumed alcohol before allowing operation. In case of unsafe conditions, the system provides alerts using a buzzer and LED indicators. Additionally, the vibration sensor helps in detecting accidents and triggering warning signals.

This project demonstrates the application of embedded systems and sensor-based technology in improving road safety. It is a simple, cost-effective, and efficient solution that can be implemented in real-life scenarios to reduce accidents and promote safe driving habits.

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CHAPTER 1 : INTRODUCTION

A) INTRODUCTION OF SMART HELMET SAFETY SYSTEM WITH ACCIDENT AND ALCOHOL DETECTION.

The **Smart Helmet Safety System with Accident and Alcohol Detection** is designed to enhance road safety by ensuring that riders follow essential safety measures. Road accidents are increasing due to negligence such as not wearing helmets and driving under the influence of alcohol.

This project uses an Arduino Uno as the main controller to monitor different safety parameters. An IR sensor is used to detect whether the rider is wearing the helmet. The MQ-3 alcohol sensor detects the presence of alcohol in the rider's breath, and a vibration sensor is used to detect sudden impacts that may indicate an accident.

If unsafe conditions are detected, the system alerts the user using a buzzer and LED indicators and disables the system using a relay. This ensures that the vehicle operates only under safe conditions. The system can also be extended with GPS and GSM modules for real-time tracking and emergency alerts.

This project demonstrates the application of embedded systems and sensortechnology in real-world safety applications, making transportation safer and smarter.

APPLICATIONS

- The system can be used in two-wheelers to ensure that riders wear helmets before starting the vehicle.
- It helps in preventing drunk driving by detecting alcohol using the MQ-3 sensor.
- It can be used to reduce road accidents by ensuring safety conditions are followed.
- The accident detection feature can be used to send emergency alerts to family members using GSM.
- It is useful for traffic safety management and enforcement of safety rules.
- The system can be implemented in smart transportation and IoT-based safety applications.
- It can be used by delivery riders and bike riders for enhanced safety

CHAPTER 2 : HARDWARE & SOFTWAR REQUIREMENT

A) Hardware Requirement

1 Arduino Uno board

Fig.2.1 Arduino Uno board

The Arduino Uno is the main controller of the system. It reads inputs from all sensors and controls the output devices like LEDs, buzzer, and relay. It processes the logic to ensure safety conditions are met.

2 IR Sensor

Fig.2.2 IR Sensor

The IR sensor is used to detect whether the helmet is worn or not. When an object (head) is near the sensor, it gives a signal to the Arduino. Based on this, the system allows or restricts operation.

3 MQ-3 Alcohol Sensor

Fig.2.3 MQ-3 Alcohol Sensor

The MQ-3 sensor detects the presence of alcohol in the rider’s breath. If alcohol is detected, it sends a signal to the Arduino. The system then activates the buzzer and disables the relay.

4 Vibration

Fig.2.4 Vibration

The vibration sensor detectssudden shocks or impacts. When a strong vibration occurs, it indicates a possible accident. The system then triggers an alert using LED and buzzer.

5 Relay Module

Fig.2.5 Relay Module

The relay module acts as a switch to control the vehicle ignition system. It turns ON only when all safety conditions are satisfied. If unsafe conditions occur, it turns OFF to stop operation

Fig.6 Buzzer

Fig.2.6 Buzzer

The buzzer is used as an alert system. It produces sound when unsafe conditions like no helmet, alcohol detection, or accident occur. It helps in warning the user immediately.

7 Green LED & Red LED

Fig.2.7 Green LED & Red LED

Green LED : The green LED indicates safe conditions. It turns ON when the helmet is worn and no alcohol is detected. It shows that the system is functioning normally.

Red LED : The red LED indicates unsafe conditions. It turns ON when the helmet is not worn, alcohol is detected, or an accident occurs. It alerts the user visually.

8 Breadboard

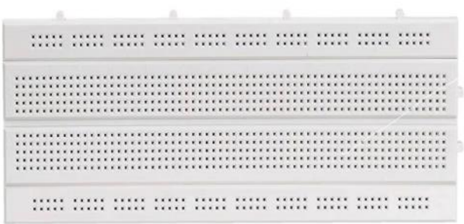


Fig.2.8 Breadboard

The breadboard is used to assemble the circuit without soldering. It helps in easy connection of components and testing. It allows flexible circuit design.

9 Jumper Wires

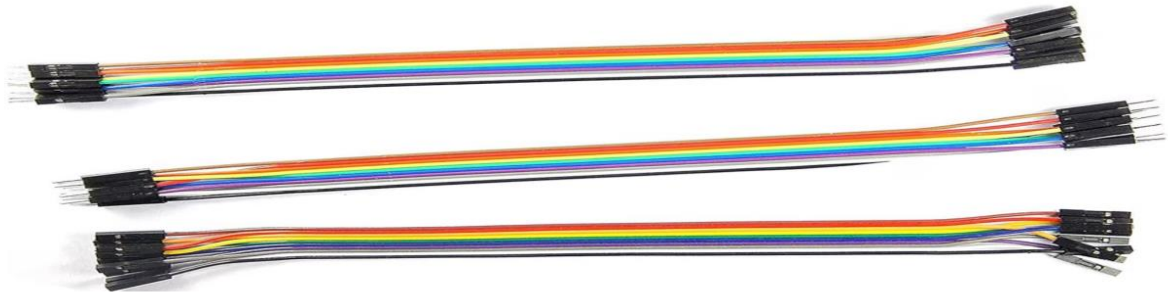


Fig.2.9Jumper Wires

Jumper wires are used to connect all components in the circuit. They provide electrical connections between sensors, Arduino, and output devices. They make the setup simple and organized.

10 Supply power

Fig.2.10 Supply power

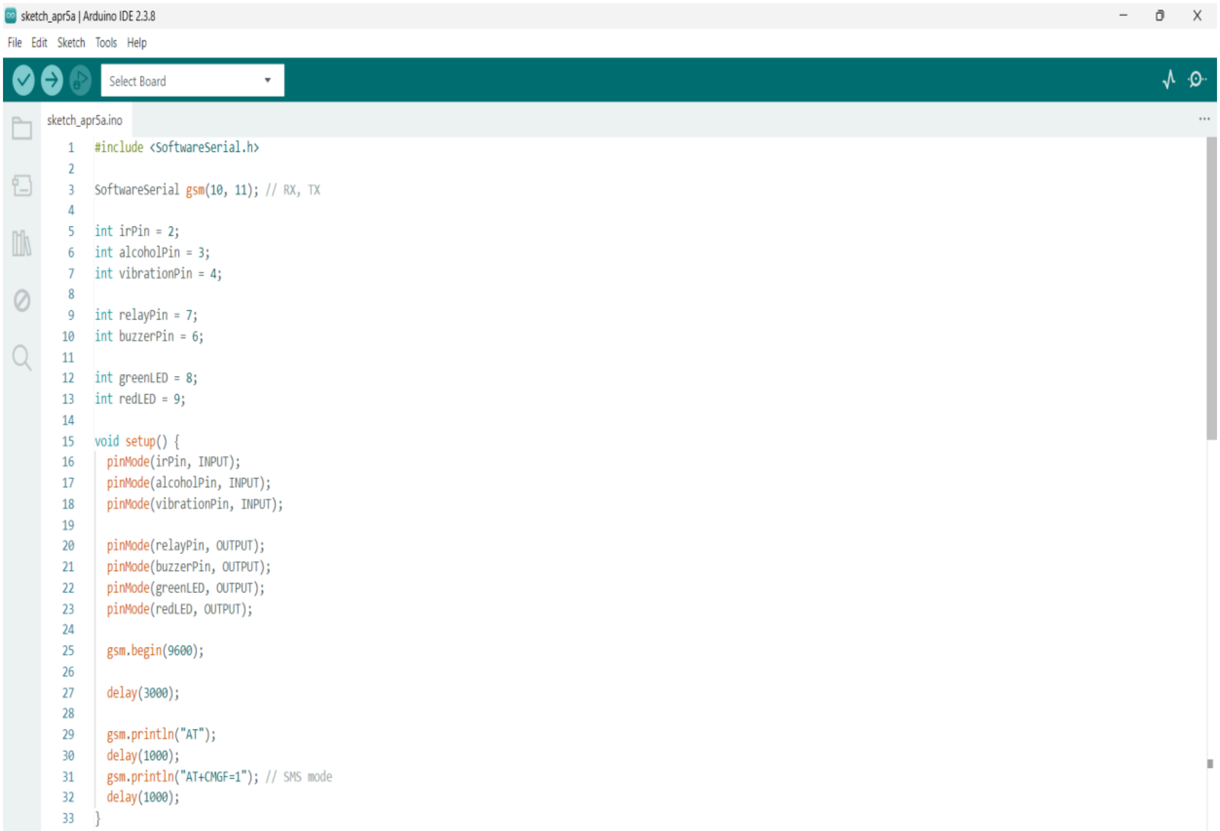
The power bank is used to provide external power supply to the Arduino Uno and connected components. It ensures stable and portable operation of the system.

B) SOFTWARE REQUIREMENTS

The software requirements for this project include the tools and platforms used to write, compile, and upload the program to the microcontroller.

Software Used : Arduino IDE

The Arduino IDE is used to write and upload the program to the Arduino Uno. It provides an easy interface for coding, compiling, and debugging. The code written in the IDE controls all sensors and output devices in the system.



CHAPTER 3.METHODLOGY

(A) Hardware Description:

The block diagram represents the overall working of the Smart Helmet Safety System. The Arduino Uno acts as the central controller of the system, which receives input signals from various sensors.

The IR sensor is used to detect whether the helmet is worn by the rider. The MQ-3 alcohol sensor detects the presence of alcohol in the rider's breath, and the vibration sensor detects sudden shocks or impacts that indicate an accident.

All these sensors send input signals to the Arduino Uno, which processes the data based on the programmed logic. If the safety conditions are satisfied, the Arduino activates the relay and turns on the green LED, indicating normal operation.

If any unsafe condition is detected, such as no helmet, alcohol consumption, or accident, the Arduino activates the buzzer and red LED and deactivates the relay to stop the system. Thus, the block diagram shows the interaction between input sensors, processing unit, and output devices to ensure rider safety.

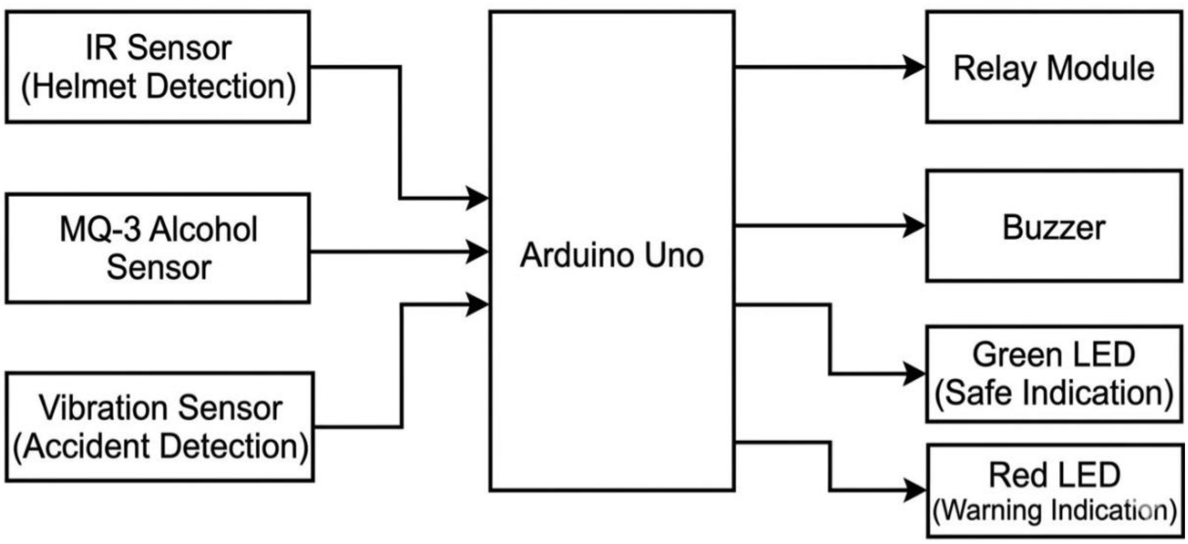


Fig.3.1 Block diagram of The Smart Helmet Safety System with Accident and Alcohol Detection.

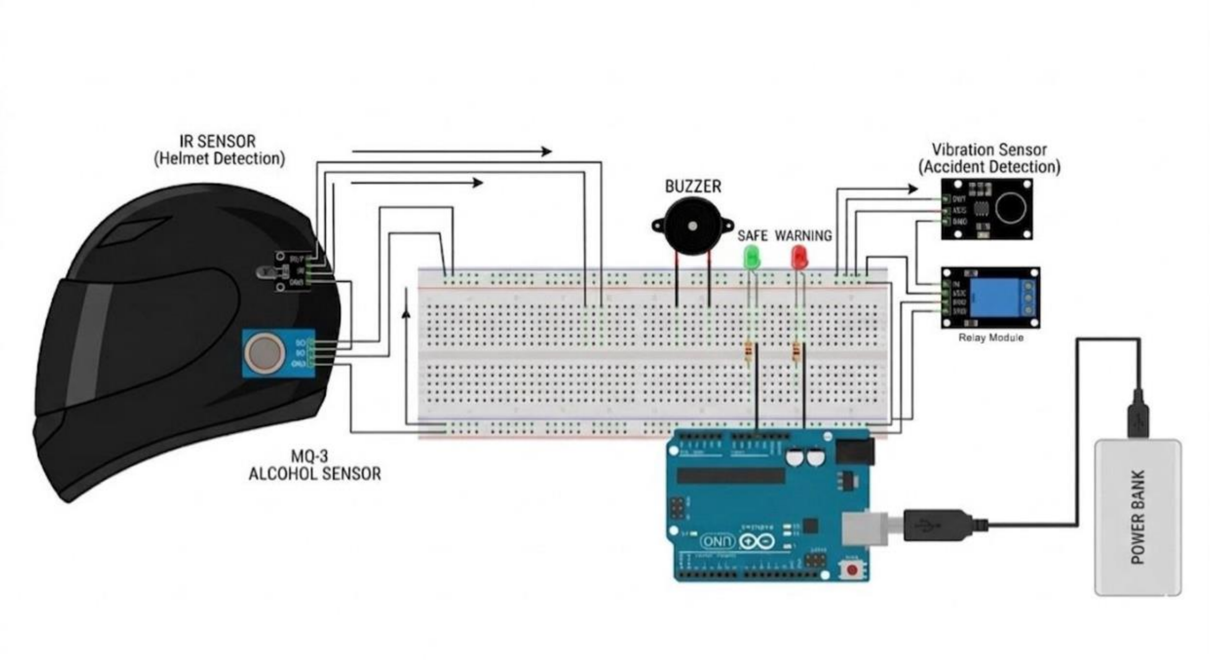


Fig.3.2 Circuit diagram of The Smart Helmet Safety System with Accident and Alcohol Detection.

CIRCUIT EXPLANATION

The circuit of the Smart Helmet Safety System is built using an Arduino Uno as the main controller. All sensors and output devices are connected to the Arduino through digital pins.

The IR sensor and MQ-3 alcohol sensor are connected to the helmet section and give input signals to the Arduino through pins 2 and 3 respectively. The vibration sensor is connected to pin 4 to detect accidents.

The output devices include a buzzer connected to pin 6, a relay module connected to pin 7, a green LED connected to pin 8, and a red LED connected to pin 9. These components provide alerts and control the system based on sensor inputs.

The GSM module is connected using pins 10 and 11 for communication. It is used to send SMS alerts in case of accidents or unsafe conditions. All components are powered using the 5V supply, and a common ground is maintained throughout the circuit.

WORKING EXPLANATION

The working of the Smart Helmet Safety System is based on input from sensors and processing by the Arduino.

Initially, the IR sensor checks whether the rider is wearing the helmet. If the helmet is worn, the system proceeds to check the MQ-3 alcohol sensor. If no alcohol is detected, the system allows operation by turning ON the green LED and activating the relay.

If the helmet is not worn or alcohol is detected, the system enters an unsafe condition. In this case, the red LED and buzzer are activated, and the relay is turned OFF to stop the system.

The vibration sensor continuously monitors for sudden shocks. If an accident is detected, the system immediately activates the buzzer and red LED. At the same time, the GSM module sends an emergency SMS alert to a predefined phone number.

Thus, the system ensures safety by checking helmet usage, preventing drunk driving, and detecting accidents in real time.

CHAPTER 4. SOFTWARE DESCRIPTION

The project is developed using Arduino IDE, which is used to write, compile, and upload the code to the Arduino Uno. The programming is done using Embedded C language, which controls the working of sensors and output devices. The software helps in testing and debugging the system easily

4.1 Source code

```
int irPin = 2;
int alcoholPin = 3;
int vibrationPin = 4;
int buzzerPin = 6;
int relayPin = 7;
int greenLED = 8;
int redLED = 9;
void setup() {
  pinMode(irPin, INPUT);
  pinMode(alcoholPin, INPUT);
  pinMode(vibrationPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(relayPin, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(redLED, OUTPUT);
}
void loop() {
  int helmet = digitalRead(irPin);
  int alcohol = digitalRead(alcoholPin);
  int vibration = digitalRead(vibrationPin);
```

```
 if (vibration == LOW) {
  digitalWrite(redLED, HIGH);
  digitalWrite(greenLED, LOW);
  digitalWrite(buzzerPin, HIGH);
  digitalWrite(relayPin, LOW);
  delay(2000); // quick alert
 }
 else if (helmet == LOW && alcohol == HIGH) {
  digitalWrite(greenLED, HIGH);
  digitalWrite(redLED, LOW);
  digitalWrite(buzzerPin, LOW);
  digitalWrite(relayPin, HIGH);
 }
 else {
  digitalWrite(greenLED, LOW);
  digitalWrite(redLED, HIGH);
  digitalWrite(buzzerPin, HIGH);
  digitalWrite(relayPin, LOW);

  delay(500); // faster response for alcohol/helmet
 }
}
```

CHAPTER 5.RESULT & DISCUSSION

5.1 Snapshot of Project

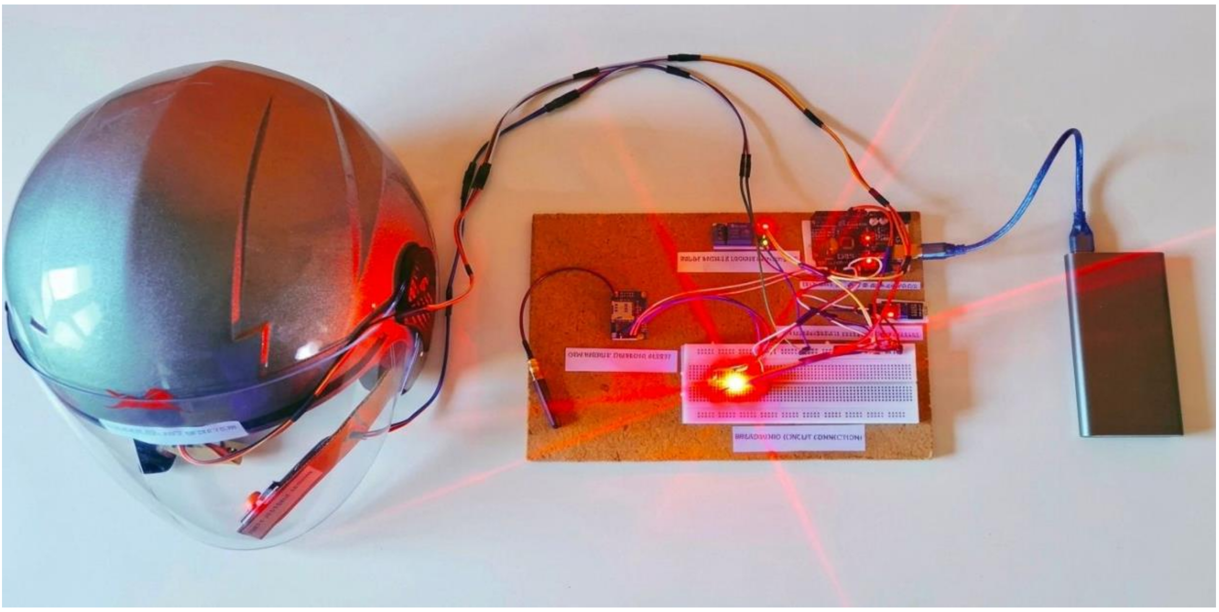


Fig.5.1 Smart Helmet Safety System with Accident and Alcohol Detection.

5.2 RESULT

The Smart Helmet Safety System was successfully designed and implemented. The system is able to detect whether the helmet is worn using the IR sensor and ensures safety conditions before operation. The MQ-3 sensor accurately detects alcohol and prevents the system from operating when alcohol is present.

The vibration sensor effectively detects sudden shocks indicating an accident and triggers an alert using buzzer and LED. The GSM module successfully sends emergency SMS alerts to the predefined mobile number. Overall, the system performs reliably and meets the intended safety objectives.

5.3 DISCUSSION

The system was tested under different conditions such as helmet detection, alcohol presence, and vibration. The results were accurate, although slight delay was observed in GSM message delivery. The project can be further improved by adding GPS tracking and better sensor calibration.

CHAPTER 6. CONCLUSION

The Smart Helmet Safety System is a simple and effective solution to improve road safety. It ensures that the rider wears a helmet and avoids driving under the influence of alcohol. The system also detects accidents and provides alerts using buzzer and GSM communication.

This project demonstrates how embedded systems and sensors can be used in real-life applications to prevent accidents and save lives. With further enhancements like GPS tracking, the system can be made more advanced and efficient.

CHAPTER 7. REFERENCES

[1] Arduino Official Website https://www.arduino.cc

[2] https://circuitdigest.com/microcontroller-projects/smart-helmet-using-arduino

[3] IR Sensor Working Principle

[4]https://www.itm ml